

EDUGENIE – GOOGLE GEMINI POWERED LEARNING ASSISTANT

PHASE 3: Project Design Phase

1. Introduction

The project design phase converts the requirements of EduGenie into a structured system plan. It describes how users interact with the system, how information moves through the application, and how the main components work together.

2. Overall System Design

The basic EduGenie design contains a user interface, input handling, AI processing, response handling, and output display. The user interface provides the interaction point while the AI service provides the generated educational response.

3. User Interaction Flow

The planned flow is: the student opens EduGenie, enters a question, submits the question, the system sends the input for AI processing, Gemini generates a response, and the response is returned and displayed to the student.

4. User Interface Design

The interface is planned to be simple and clean. A clear input area, interaction control, response area, and basic project information are included. The design gives priority to readability and ease of navigation.

5. Input Design

Input design focuses on allowing students to enter natural-language questions without complicated commands. The system should accept ordinary educational questions and pass the relevant input to the AI processing stage.

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PHASE 3: Project Design Phase – CONTINUED

6. Output Design

Responses should be presented in a readable format. Important information should be separated clearly so students can understand the answer without unnecessary visual complexity.

7. Gemini Integration Design

The AI component is designed around Google Gemini. The application sends the user's question through the required integration method and receives the generated response for display. The exact implementation depends on the chosen development environment.

8. Error and Exception Design

The design considers possible problems such as empty input, network failure, unavailable AI service, or an unsuccessful response. The application should provide understandable feedback rather than leaving the user confused.

9. Security and Responsible Use

The design should avoid collecting unnecessary personal information. AI responses should be treated as assistance rather than unquestionable facts, especially for academic or important information.

10. Conclusion

The design phase provides a blueprint for development. By defining the workflow, interface, AI integration, input, output, and error handling before coding, the team can develop EduGenie in a more organized manner.